



# Open manifold microchannel heat sink for high heat flux electronic cooling with a reduced pressure drop

Nicholas Gilmore, Victoria Timchenko, Chris Menictas\*

School of Mechanical and Manufacturing Engineering – UNSW, Sydney 2052, Australia

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## ABSTRACT

To meet the intensifying demands of electronics cooling the current study proposes an open manifold microchannel heat sink, which substantially reduces the pressure drop of a conventional manifold microchannel heat sink while retaining a low thermal resistance, thin profile ( $< 2$  mm) and planar geometry suited to microfabrication. The open heat sink achieves this by eliminating ineffective nozzle constrictions, specifically by removing channel walls beneath the slot nozzle. Two three-dimensional conjugate heat transfer models are used, including a single microchannel or a single manifold channel pair with hundreds of interconnecting microchannels. The performance of the open geometry is compared to an unmodified baseline geometry. In the microchannel model, the open case showed ideal pressure drop reductions of 45 % to 75 % with a negligible change to thermal resistance, at flow rates between 0.2 L/min and 1.0 L/min. Chamfering the tips of the microchannel walls show similar trends, although to a lesser degree. In the manifold model, the open case showed pressure drop reductions of 25 %, with thermal resistance slightly decreasing at flow rates below 0.58 L/min ( $-4$  % to  $1$  %) and increasing at flow rates above 0.70 L/min ( $11$  % to  $21$  %). These changes in thermal resistance were attributed to flow maldistribution, which may be alleviated through optimisation of the manifold convergence profile. At a flow rate of 1.32 L/min, the open geometry achieved a thermal resistance of  $0.120$   $\text{cm}^2\text{K/W}$  with a pressure drop of  $10.3$  kPa.

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## 1. Introduction

There is an increasing demand for high-performance electronics. Data centres already consume about 2 % of the US power generation, 40 % of this from requisite cooling systems [1]. This consumption has significant economic and environmental impacts. In the United States, data centre energy usage costs \$13 billion each year with 150 million tons of associated carbon emissions [2]. A shift from air to liquid cooling may improve energy efficiencies. Liquid cooling can achieve higher heat transfer rates, which allow target chip temperatures ( $< 85$  °C) even with high coolant inlet temperatures ( $70$  °C) [3]. High inlet temperatures may reduce data centre power consumption by as much as 50 %, through the reduced use of chillers, which air systems rely upon [3,4]. Higher coolant outlet temperatures have more utility for secondary applications such as space heating or cooling, water heating, power generation or desalination [5]. Liquid cooling is generally preferred to air cooling for waste heat recovery, due to superior thermophysical properties. Continual miniaturisation of electronics may exceed the

cooling limits of air, around  $37$   $\text{W/cm}^2\text{K}$  [6]. Liquid heat exchangers can achieve smaller form factors, which is crucial for emerging architectures like 3D chips [7,8]. For these high heat flux cooling applications, there are several emerging single and two-phase liquid cooling methods, including microchannel, jet impingement and immersion.

Tuckerman and Pease first introduced microchannels in 1981 [9]. The microchannel heat sink achieved a high heat transfer coefficient of  $11.1$   $\text{W/cm}^2\text{K}$ , with a substantial pressure drop of around  $214$  kPa. Since then, numerous studies have sought to increase microchannel heat transfer rates while decreasing pressure drop. Review papers address these studies well [10–13]. Manifold microchannels substantially reduce the pressure drop. Conventional designs consist of two layers, a top layer of interdigitated manifold channels and a bottom layer of microchannels. The obliquely orientated microchannels interconnect pairs of inlet and outlet manifold channels. The manifold channels have a large diameter to reduce the incurred pressure drop, while the microchannels have a narrow diameter to enhance heat transfer rates.

Escher et al. [3,14] numerically and experimentally optimised a conventional manifold microchannel heat sink, achieving a heat transfer coefficient of  $11.1$   $\text{W/cm}^2\text{K}$  with a pressure drop of  $22$  kPa.

\* Corresponding author.

E-mail address: [c.menictas@unsw.edu.au](mailto:c.menictas@unsw.edu.au) (C. Menictas).

## Nomenclature

|               |                                   |
|---------------|-----------------------------------|
| $C$           | heat capacity ( $J/kg K$ )        |
| $d$           | hydraulic diameter ( $m$ )        |
| $F$           | body force vector ( $N/m^3$ )     |
| $f$           | friction factor                   |
| $h$           | height ( $m$ )                    |
| $K$           | number of manifold pairs          |
| $k$           | thermal conductivity ( $W/m K$ )  |
| $\dot{m}$     | mass flow rate ( $kg/s$ )         |
| $N$           | number of microchannels           |
| $p$           | pressure ( $Pa$ )                 |
| $Q$           | heat flux ( $W/cm^2$ )            |
| $R$           | thermal resistance ( $cm^2 K/W$ ) |
| $Re$          | Reynolds number                   |
| $T$           | temperature ( $K$ )               |
| $\mathbf{u}$  | velocity vector ( $m/s$ )         |
| $V$           | flow rate ( $L/min$ )             |
| $v$           | velocity ( $m/s$ )                |
| $w$           | width ( $m$ )                     |
| $xyz$         | coordinates ( $m$ )               |
| $X$           | placeholder variable              |
| $\mathcal{L}$ | characteristic length             |

## Greek Letter

|          |                                 |
|----------|---------------------------------|
| $\Delta$ | difference / Drop               |
| $\rho$   | density ( $kg/m^3$ )            |
| $\mu$    | dynamic viscosity ( $Pa s$ )    |
| $\Pi$    | constriction or expansion ratio |
| $\tau$   | viscous stress tensor           |
| $\nu$    | kinematic viscosity ( $m^2/s$ ) |

## Subscript / Superscript

|     |                             |
|-----|-----------------------------|
| %   | proportional percentage     |
| 1/2 | half of                     |
| amb | ambient                     |
| app | apparent                    |
| b   | base                        |
| chf | corner chamfer              |
| Ch  | channel                     |
| f   | fluid                       |
| HT  | heat transfer, microchannel |
| in  | inlet / In                  |
| L   | length                      |
| m   | manifold                    |
| max | maximum                     |
| N   | nozzle                      |
| p   | constant pressure           |
| tot | total                       |
| vd  | viscous dissipation         |
| w   | wall                        |

Paredes et al. [15] tested the heat sink on a highly concentrated photovoltaic system. At 955 suns, or around  $100 W/cm^2$ , with an elevated water inlet temperature of  $79.3^\circ C$ , the heat sink could maintain required cell temperatures and output fluid at a useful temperature of around  $81.8^\circ C$ . The heat sink had a large area of  $53 mm \times 57 mm$  and a thin profile of less than  $2 mm$ . Across the flow rates of  $5 L/min$  to  $15 L/min$ , the pressure drops ranged from  $3 kPa$  to  $17 kPa$ . A large portion of this pressure drop came from the heat sink's slot nozzles. These nozzles promote flow distribution across the microchannels and also produce a jet. This jet enhances local heat transfer rates in the impingement zone, while also decreasing them in the recirculation region. As noted by the study, these counteracting effects limit the net heat transfer bene-

fits of increasing the jet velocity. As such, there is an opportunity to reduce the pressure drop associated with the nozzle constriction while still maintaining thermal performance.

Han et al. [16] investigated a manifold microchannel heat sink with circular jet nozzles, using drainage trenches to mitigate cross-flow effects. The heat sink achieved a high heat transfer coefficient of  $20.0 W/cm^2 K$ . However, it incurred a high-pressure drop of  $28 kPa$ , with a flow rate of  $0.5 L/min$  and over a small  $8 mm \times 8 mm$  area. Colgan et al. [17] also utilised circular jet nozzles in a manifold microchannel heat sink with nozzles positioned at either end as opposed to being interspaced over channels. The heat sink achieved a moderate heat transfer coefficient of  $9.5 W/cm^2 K$ , with a high-pressure drop of  $32.4 kPa$ . The following studies present other manifold microchannel heat sink cases [18–20]. Collectively, these studies provide a robust baseline for performance comparisons. They also highlight alternative uses of jet impingement in manifold microchannel heat sinks.

In their review, Siddiqui and Zubair [21] demonstrate the importance of inlet and outlet connections for manifold flow uniformity and pressure drop. Variables including location, orientation, shape and number. The current study assumes a single lateral inlet and outlet connection. This arrangement is often convenient for electronic applications [3]. Ramos-Alvarado et al. [22] tested various channel configurations, also aiming to promote flow uniformity across the manifold. The Parallel-II-A and Parallel-II-B configurations were most akin to a manifold microchannel heat sink. These configurations had channel flow rates  $0.76 \times$  to  $1.23 \times$  the average flow rates across all channels. It is necessary to consider similar flow non-uniformity when validating the current model with experimental results from Escher, et al. [3].

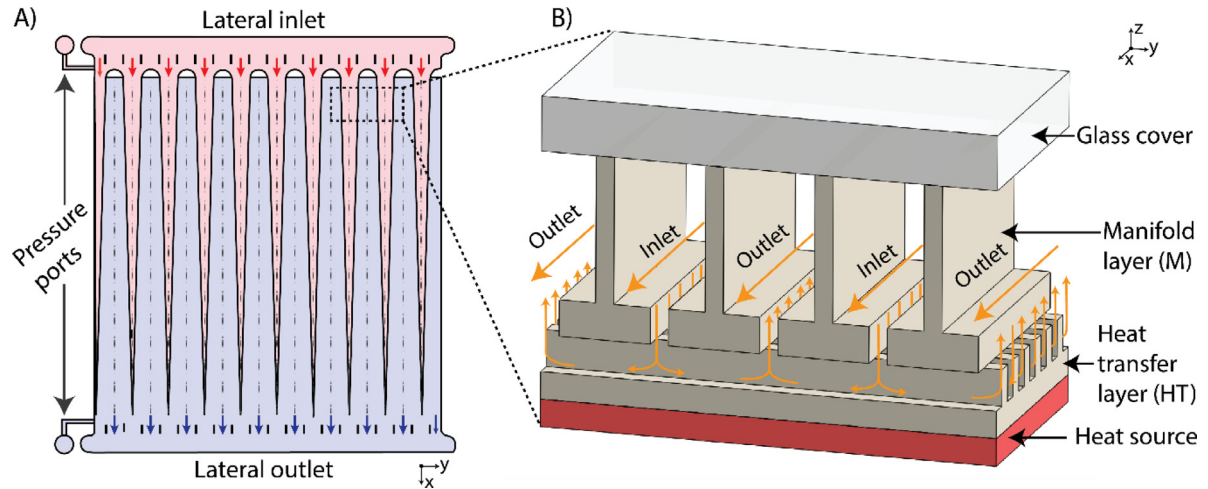
Manifolds use converging inlets and diverging outlets to achieve uniform flow rate across microchannels, tapering either the height or width [3,20]. For width tapered manifolds, the slot nozzle width determines the minimum manifold channel width. Ryu et al. [23] studied the influence of slot nozzle width in their parametric study. At high flow rates and heat fluxes, manifold microchannel heat sinks are susceptible to flow maldistribution across microchannels. The current study adopts the profile developed by Escher et al. [3,20]. Dedicated optimisation of this profile would improve flow distribution. However, this is not feasible given the computational demands of the current model.

Various studies have investigated a range of channel paths, shapes or other features. Examples include wavy channels [24], flow disruptors [25] or herringbone [26]. In most cases, these improve heat transfer rates for some pressure drop penalty. For simplicity, the current study investigates straight rectangular channels. Although, these design features do present scope for future work.

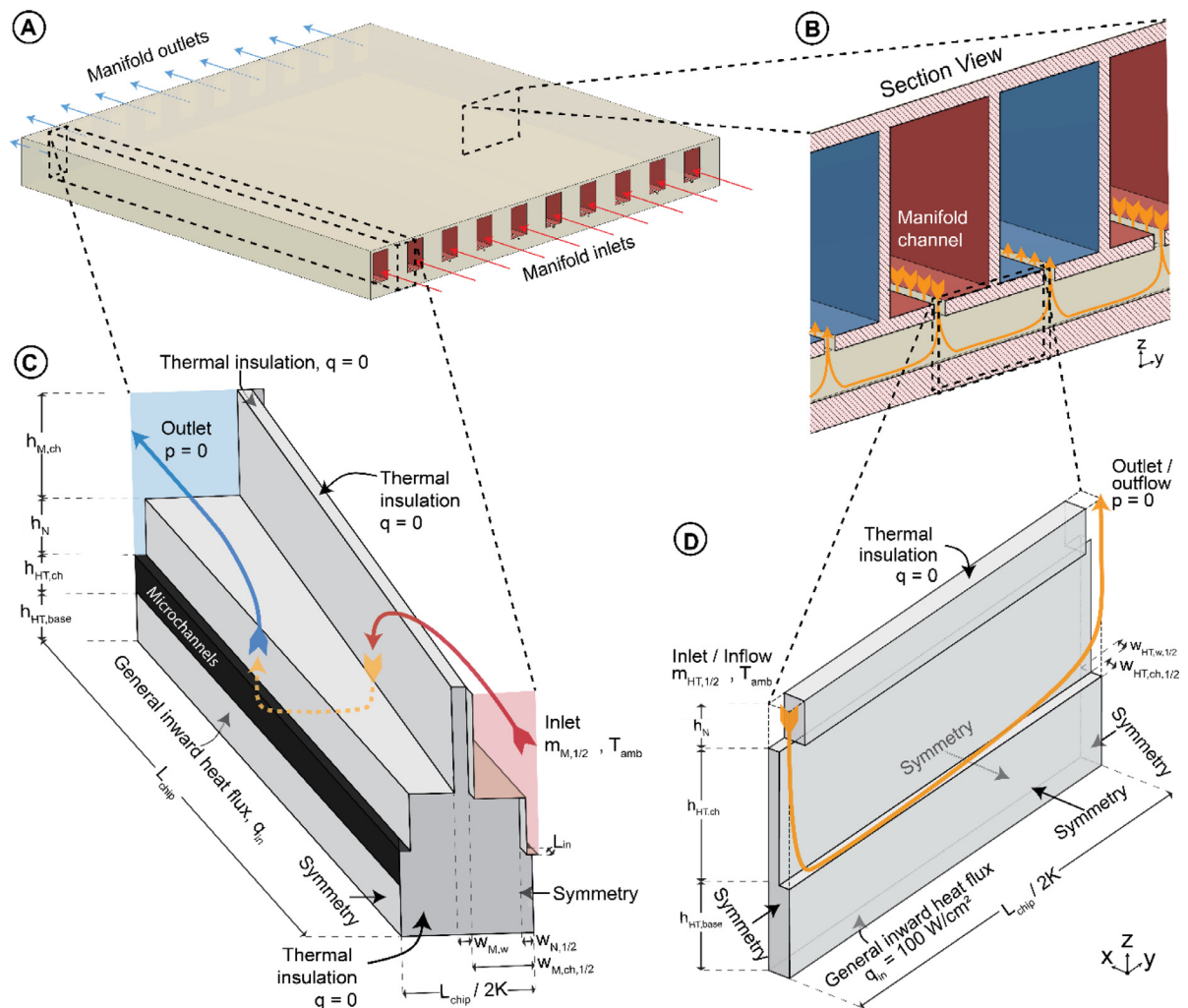
By eliminating ineffective nozzle constrictions, the current study aims to reduce the pressure drop of a conventional manifold microchannel heat sink while retaining thermal performance and simple planar geometry. The manifold microchannel heat sink presented by Escher et al. [3,14] serves as the baseline case. The study uses two conjugate heat transfer models. The first includes a single microchannel, the second a single manifold channel with hundreds of interconnecting microchannels. Section 2 describes each model including mesh independence and validation, Section 3 presents and discusses the results and and Section 4 summarises key findings and identifies areas for future work.

## 2. Methodology

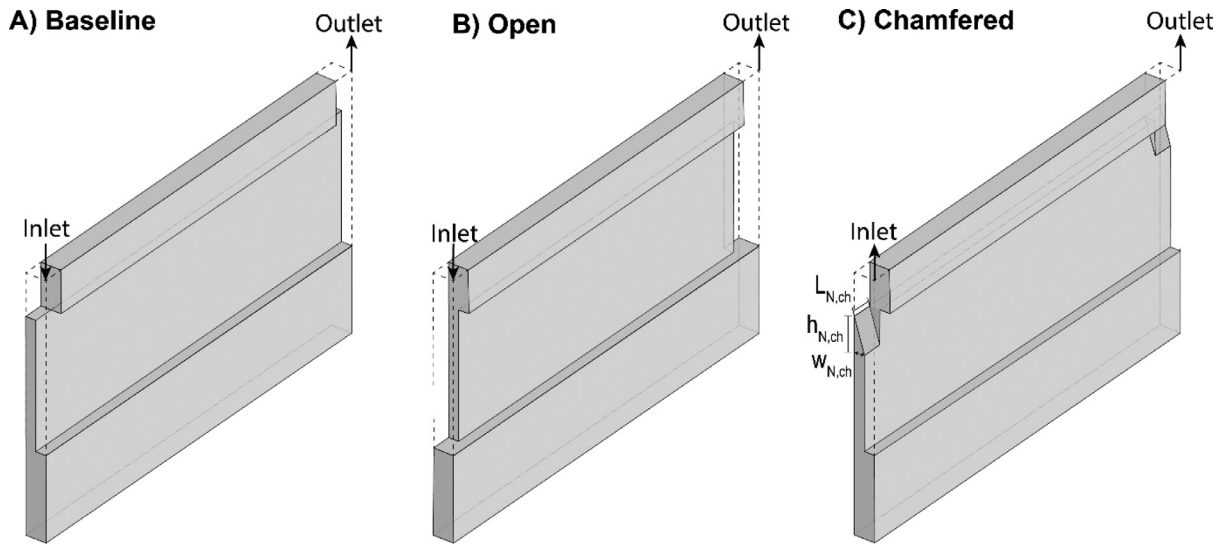
This study considers a conventional manifold microchannel heat sink, as shown in Figs. 1 and 2. The heat sink consists of two layers, an upper layer of manifolds channels and a lower layer of microchannels. Inlet and outlet manifold channels are interdigitated with each other, running inwards from opposite sides of the heat



**Fig. 1.** A) Top view of the manifold microchannel heat sink that was experimentally studied by Escher et al. [3]. Note the converging-diverging manifold channels and the positioning of the pressure ports next to the half-width inlet manifold channel. B) Cross-section of a conventional manifold microchannel heat sink where microchannel walls run continuously across the heat sink.



**Fig. 2.** Combined schematics for numerical models with baseline geometries. A) Total heat sink structure. B) Detailed cross-section of manifold channels. C) Manifold model geometry and boundary conditions. D) Microchannel model geometry and boundary conditions.



**Fig. 3.** Baseline and proposed geometries shown in scope of microchannel model A) Baseline, B) Open and C) Chamfered, where chamfer height ( $h_{N, chf}$ ) was varied: 12.5  $\mu\text{m}$ , 25  $\mu\text{m}$ , 50  $\mu\text{m}$ , 75  $\mu\text{m}$ , 100  $\mu\text{m}$ , 150  $\mu\text{m}$ , 200  $\mu\text{m}$ , 250  $\mu\text{m}$  and 300  $\mu\text{m}$ . B) Channel walls removed, or opened, directly beneath the slot nozzles.

sink. Microchannels run laterally underneath, connecting the inlet and outlet manifold channels. The manifold channels have a large diameter to distribute flow with minimal pressure drop. They also have a tapered profile to promote uniform microchannel flow rate. The microchannels have a narrow diameter to enhance heat transfer rates. The flow enters and exits the microchannels via a narrow slot nozzle in the base of the manifold channels. This nozzle forms a jet which enhances local heat transfer rates and increases flow distribution across the microchannel length. However, this narrow nozzle also causes significant pressure losses. To mitigate this, the current study proposes the open and chamfered geometries shown in Fig. 3.

### 2.1. Description, geometry and boundary conditions

This study uses two three-dimensional conjugate heat transfer models. The microchannel model simulates a single microchannel, while the manifold model simulates a pair of inlet-outlet manifold channels. The microchannel model is used to investigate a range of geometries. Whereas the manifold model only investigates a single geometry, originally optimised by Escher et al [3]. Other variations aren't investigated by the manifold model due to the computational demands of simulating the hundreds of interconnecting microchannels. Fig. 2 provides schematics for each model, while Table 1 provides all parameters used for the models.

The microchannel model tests at three microchannel widths: 25  $\mu\text{m}$ , 50  $\mu\text{m}$  and 100  $\mu\text{m}$ . The widths of the microchannel walls aligning with these values. Conversely, the manifold model tests at a microchannel width of 26.98  $\mu\text{m}$ , and wall width of 11.98  $\mu\text{m}$ . Both cases keep the total height below 2 mm, the thin profile desirable for electronic cooling. The following expression determines the tapering of the manifold wall, developed by Escher et al. [3].

$$w_{M, ch}^{in}(x) = w_{M, ch, 0}^{in} (7.2 \times 10^8 x^5 m^{-5} - 3.25 \times 10^7 x^4 m^{-4} + 6.6 \times 10^5 x^3 m^{-3} - 8.14 \times 10^3 x^2 m^{-2} + 1.45 x m^{-1} + 1) \quad (1)$$

As it enters the heat sink, water divides across the inlet manifold channels. It then divides across the hundreds of microchannels to reach the outlet manifold channels. If we assume uniform flow distribution and account for symmetry, then the inlet mass flow rate for the microchannel model is:

$$\dot{m}_{HT, 1/2} = \frac{\dot{V}_{tot}}{4 \cdot K \cdot N} \cdot \rho_f \quad (2)$$

Where  $\dot{V}_{tot}$  is the total flow,  $K$  is the number of manifold channels,  $N$  is the number of microchannels and  $\rho_f$  is the water density. Straight manifold channels can cause up to a 70 % difference in flow rate, across the individual microchannels. Microchannel velocities decrease in the streamwise direction as the volume flux decreases. Tapered channels can counteract this through a corresponding decrease in cross-sectional area. That is converging inlet and diverging outlet manifold channels. Modelling of this may bring flow rates within 4 % [14].

Assuming a uniform flow distribution, the manifold flow rate is:

$$\dot{m}_{M, 1/2} = \frac{\dot{V}_{tot}}{K} \cdot \rho_f \quad (3)$$

For comparing the baseline and open cases, this uniformity is reasonable. However, validation needs to consider flow maldistribution present in experiments. Both models calculate the pressure drop as the difference between the pressure at the inlet and outlet boundaries. The heat sinks total pressure drop is the summation of the pressure drop across the manifold layer and the heat transfer layer.

$$\Delta p_{tot} = \Delta p_M + \Delta p_{HT} \quad (4)$$

The manifold model captures this total pressure drop, while the microchannel model only captures pressure drop across the slot nozzles and microchannels. This pressure drop is described by the following expression developed by Escher et al. [14]:

$$\Delta p_{HT} = 4f_{app, N} \frac{\rho_f u_N^2 L_N}{d_N} + 2f_{app, ch} \frac{\rho_f u_{ch}^2 L_{ch}}{d_{ch}} + \Pi_{in} \frac{\rho_f u_N^2}{2} + \Pi_{out} \frac{\rho_f u_{ch}^2}{2} \quad (5)$$

Terms respectively account for pressure drop across the nozzle length, microchannel length, inlet nozzle constriction and outlet nozzle expansion. The following expression calculates the heat sink total thermal resistance:

$$R_{tot} = \frac{T_{max} - T_{f, in}}{q} \quad (6)$$

Where  $T_{max}$  is the maximum heat sink temperature,  $T_{f, in}$  is the temperature of the inlet fluid and  $q_{in}$  is the applied heat flux. The total thermal resistance consists of three components: bulk, convective and conductive. The conductive thermal resistance accounts

**Table 1**

Parameters used in geometry, material and boundary conditions for microchannel and manifold models.

| Description                         | Symbol        |        |                   | Microchannel model | Manifold model    |
|-------------------------------------|---------------|--------|-------------------|--------------------|-------------------|
| Height of heat transfer channel     | $h_{HT, ch}$  | 300    | $\mu m$           | 226                | $\mu m$           |
| Height of heat transfer base        | $h_{HT, b}$   | 200    | $\mu m$           | 525                | $\mu m$           |
| Height of heat transfer nozzle      | $h_{HT, N}$   | 100    | $\mu m$           | 350                | $\mu m$           |
| Height of heat transfer, total      | $h_{HS, tot}$ | 2000   | $\mu m$           | 2000               | $\mu m$           |
| Height of manifold channel          | $h_{M, ch}$   | -      | -                 | 650                | $\mu m$           |
| Width of heat transfer nozzle       | $w_{HT, N}$   | 100    | $\mu m$           | 133                | $\mu m$           |
| Width of the heat transfer wall     | $w_{HT, w}$   | 50     | $\mu m$           | 11.98              | $\mu m$           |
| Width of the heat transfer channel  | $w_{HT, ch}$  | 50     | $\mu m$           | 26.98              | $\mu m$           |
| Width of manifold wall              | $w_{M, w}$    | -      | -                 | 87                 | $\mu m$           |
| Length of the heat transfer channel | $L_{HT, ch}$  | 1000   | $\mu m$           | 800                | $\mu m$           |
| Length of heat sink chip            | $L_{chip}$    | -      | -                 | 20000              | $\mu m$           |
| Length of manifold inlet            | $L_{in}$      | -      | -                 | 500                | $\mu m$           |
| Number of manifold channel pairs    | $K$           | 10     | -                 | 12.5               | -                 |
| Number of microchannels             | $N_{HT, ch}$  | -      | -                 | 514                | -                 |
| Inlet temperature                   | $T_{in}$      | 293.15 | K                 | 293.15             | K                 |
| Silicon thermal conductivity        | $k_s$         | 149    | W/(m · K)         | 130                | W/(m · K)         |
| Total volumetric flow rate          | $V_{tot}$     | 1.00   | L/min             | 0.2→1.32           | L/min             |
| Heat flux                           | $Q$           | 100    | W/cm <sup>2</sup> | 100                | W/cm <sup>2</sup> |

for conduction up the heat sink's base and walls. The model assumed a silicon material, which suits microfabrication techniques, can eliminate thermal contact resistances and avoid damage from thermal expansion. Both models use a constant thermal conductivity. For the microchannel model, the conductivity is 149 W/(m · K) matching the value used in the numerical study Escher et al. [14]. For the manifold model, the conductivity is 130 W/(m · K). This value matching the default software value to align with the experimental validation case [3]. The lower conductivity provides a more conservative thermal resistance value.

For the microchannel model, the top boundary condition is set to adiabatic. This condition will cause the microchannel model to slightly overestimate the heat sinks overall thermal resistance. This is because this top boundary adjoins to the manifold channels, which contribute to the heat sinks total thermal resistance. The degree of this contribution is captured by the manifold model, which simulates both microchannels and manifold channels. For the manifold model, the top boundary condition is also set to adiabatic. In the experiments used for validation this top boundary is insulated by a glass layer, which reduces operational heat loss to about 1.5 % [3]. Some of this heat loss may also occur from the sides of the heat sink. Thus, the adiabatic boundary condition is acceptable.

The study proposes two geometries, referred to as chamfered and open. These geometries are illustrated in Fig. 3. Both geometries aim to reduce the pressure drop caused by the constriction and expansion at the slot nozzle tip. The open geometry achieves this by removing channel walls from beneath the slot nozzle tip, which effectively doubles the minimum cross-sectional area depending on the ratio of the channel and wall widths. The performance of these proposed geometries are compared to the baseline geometry as follows:

$$X_{\%} = \frac{X_{base} - X_{open}}{X_{open}} \quad (7)$$

Where  $X_{base}$  is the baseline geometry value and  $X_{open}$  is the value for the proposed geometry. Both models use COMSOL Multiphysics 5.5. The microchannel model was run on a local computer with an Intel i7-8750H CPU at 2.20GHz with 32 GB of available memory. The manifold model used a geometric multigrid solver, run remotely utilising two Intel Xeon Gold 6144 CPU at 3.50 GHz and with 772.43 GB of available memory.

## 2.2. Assumptions and governing equations

The conjugate heat transfer problem was solved using the conservation of mass, momentum and energy in the fluid domain

and conservation of energy in the solid domain. The liquid flow is assumed to be incompressible, given the low pressures involved. The model assumed temperature-dependent fluid properties as fluid temperatures significantly increase across the short channel length [28,29]. The model also assumes viscous dissipation, which may very slightly increase the fluid temperature due to the high-velocity gradients occurring in the narrowest channels [27]. The conservation of mass, or continuity equation is:

$$\rho \nabla \cdot \mathbf{u} = 0 \quad (8)$$

Where  $\rho$  (kg/m<sup>3</sup>) is the density and  $\mathbf{u}$  (m/s) is the velocity vector. The conservation of momentum, or Navier-Stokes equation:

$$\rho (\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla \cdot [-\rho \mathbf{I} + \mu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] + \mathbf{F} \quad (9)$$

Where  $\mu$  (Pa · s) is the dynamic viscosity and  $\mathbf{F}$  (N/m<sup>3</sup>) is the body force vector. The conservation of energy equation:

$$Q_{vd} = \rho C_p \mathbf{u} \cdot \nabla T + \nabla \cdot \mathbf{q} \quad (10)$$

$$\mathbf{q} = -k \nabla T \quad (11)$$

$$Q_{vd} = \tau : \Delta \mathbf{u} \quad (12)$$

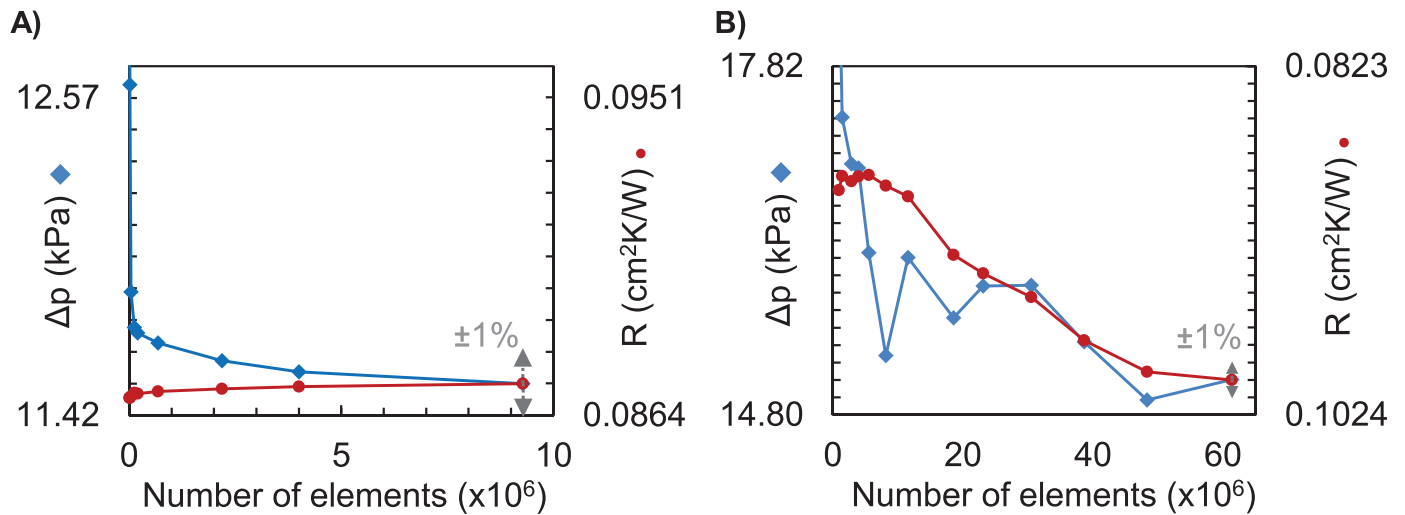
Where  $Q_{vd}$  (W/m<sup>3</sup>) is the viscous heating term,  $C_p$  (J/(kg · K)) is the specific heat capacity at constant pressure,  $T$  (K) is the absolute temperature,  $\mathbf{q}$  (W/m<sup>2</sup>) is a heat flux defined by Fourier's Law of heat conduction,  $k$  (W/(m · K)) is the thermal conductivity and  $\tau$  (Pa) is the viscous stress tensor. The model applies a no-slip boundary condition at the fluid-solid interface.

The flow was assumed to be laminar. A peak Reynolds Number of 70 and 1700 were calculated for the microchannel and manifold models, respectively. This less than the transitional range of around 2500. The Reynolds number calculated with the expression from Muzychka and Yovanovich [30]:

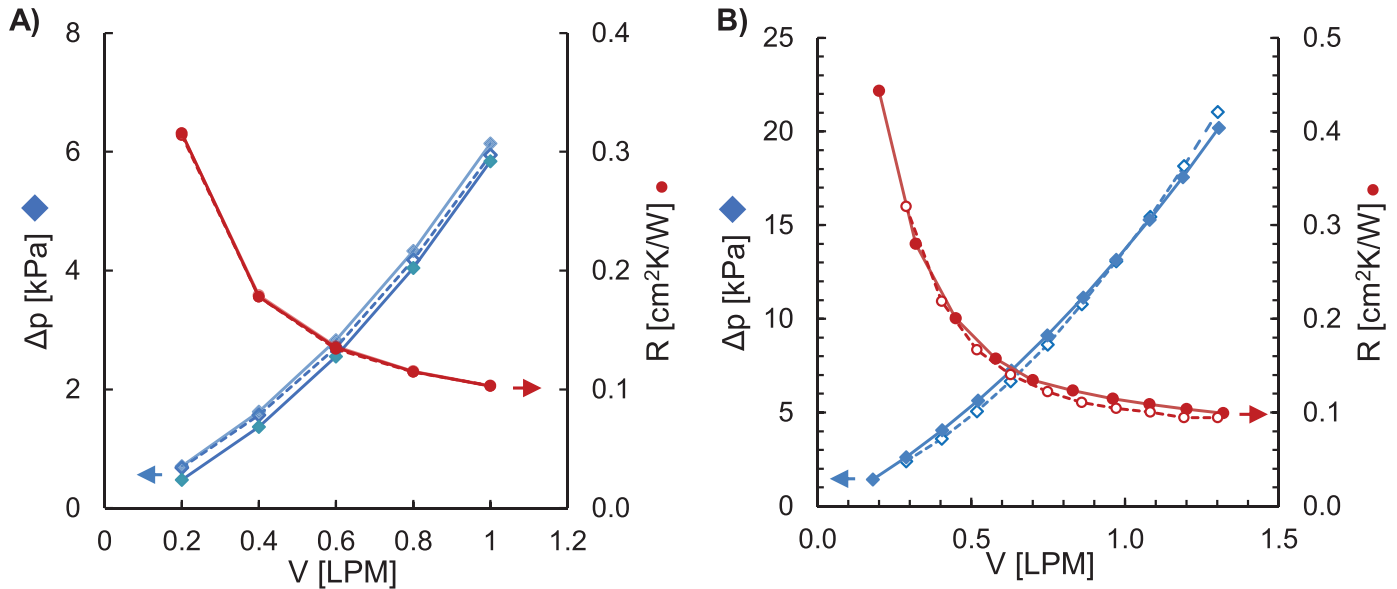
$$Re = \frac{v_{avg} \mathcal{L}_{\sqrt{A}}}{\nu} \quad (13)$$

## 2.3. Mesh

The model used unstructured tetrahedral mesh for all domains. The fluid domain used a finer mesh, with refinement used at boundaries and corners and portioning of domains to improve mesh quality. The microchannel model used increasingly fine meshes, between 11,189 and 9,279,918 elements. The model was tested at a flow rate of 1 L/min and with narrow channels,



**Fig. 4.** Mesh independence study for the A) microchannel model B) manifold model. The y-axis increments sized at 1 % of the value obtained by the finest mesh to clearly show proportionate changes.



**Fig. 5.** Validation results for A) microchannel model and B) manifold model. Solid lines are results for the current models, while dashed lines are results from the validation cases [3,14]. The lighter blue line shows the microchannel case with constant fluid properties.

$w_{HT, ch} = 25 \mu\text{m}$ , to ensure a more robust convergence. The thermal resistance converged quickly, with the result for the coarsest mesh being within 0.5 % of the finest mesh result. At 2,193,332 elements, the pressure drop value was within 0.7 % of the finest mesh result, requiring only 10 % of the computational time. This accuracy is, therefore, acceptable for the study. The manifold model used meshes between 940,500 and 60,376,263 elements with tests run at the highest flow rate of 1.32 L/min. At 38,674,042 elements, thermal resistance and pressure drop values were within 3.3 % and 1.8 % of the finest mesh, at 48,323,022 elements they were within 0.5 % and 1.2 % of the finest mesh at 61,376,263 elements. As a result of this convergence study, the finest mesh was used for obtaining results. Fig. 4 plots the results of the mesh convergence study.

#### 2.4. Validation and verification

Numerical results from Escher et al. [14] were used for validation of the microchannel model. The comparison of the current

model and validation data is presented in Fig. 5A. The thermal resistance was within  $0.0018 \text{ cm}^2\text{W/K}$  or -0.2 % to 0.9 % of validation data, pressure drop was also within 0.20 kPa or -29.8 % to -1.8 %. Different assumptions used by the current and validation model account for the slight underprediction of pressure drop. The current model assumed temperature-dependent fluid properties, whereas the earlier model assumed constant fluid properties. The fluid temperature increases significantly in the streamwise direction, which causes a decrease in its viscosity. For example, at 0.2 L/min, the fluid's temperature rises by  $29^\circ\text{C}$ , which corresponds to a 44 % decrease in fluid viscosity. The current model was run without temperature-dependent fluid properties to demonstrate the effect of this assumption. Thermal resistance was unaffected by the assumption; however, the pressure drop increased to within 0.19 kPa or 4.7 % of the validation case.

Experimental results from Escher et al. [3] were used to validate the manifold model with the values compared in Fig. 5B. Pressure drop values were within 1 Pa or -5 % to 12 % of the validation case, thermal resistance values also within  $0.012 \text{ cm}^2\text{K/W}$  or -4 %

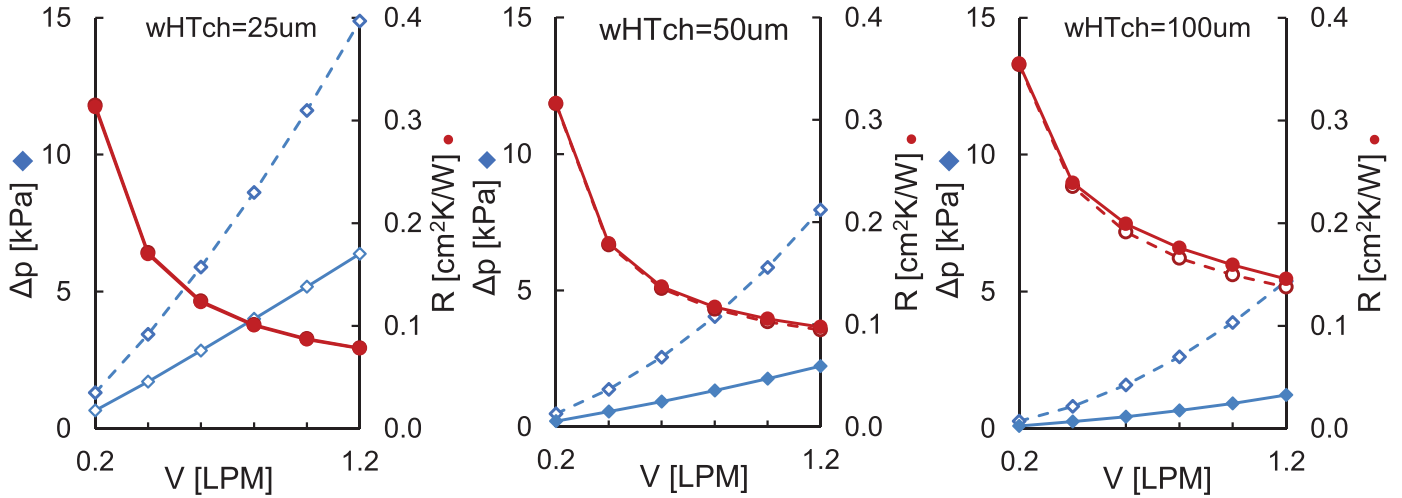


Fig. 6. Pressure drop and thermal resistance for the baseline (dashed line) and open channel (solid line) geometries.

to 9 %. Therefore, this accuracy is reasonable for this study. Validation experiments positioned pressure ports at the edge of the heat sink, next to the half-width inlet manifold channel as shown in Fig. 1. Recall, there are twelve other full-width manifold channels. As a result, the half-width channel will exhibit a higher pressure drop and a lower flow rate than the twelve other full width channels. Pressure drop was measured adjacent to this half width channel, so the model must account for this. A no-slip boundary condition was applied at the inlet manifold channel, to simulate the half channel width. This condition increased the pressure drop by  $1.9 \times$  at 0.2 L/min and  $1.3 \times$  at 1.2 L/min. It was then necessary to correct the flow rate as the higher pressure drop would result in non-uniform flow distribution. It is difficult to calculate this value without modelling the entire heat sink, which is not feasible. As such, a fitting value of 0.9 was estimated as follows:

$$v_{non-uniform} = \frac{0.9 \times \dot{V}}{w_{HTch} h_{HTch} K} \quad (14)$$

This estimation provides a good agreement between the model and experimental results and fits findings from previous studies on flow maldistribution in manifolds [22]. The experimental heat sink had other features which may have influenced the validation results, although these features will not influence comparisons between the baseline and proposed geometries. The experimental heat sink included two small bracing ribs, for structural support. It also had two narrow strips which exhibited low heat flux input, due to being blocked by temperature sensor. The previous study already addressed the effects of these features. As such, this model omits them to simplify the already computationally demanding model. The validation experiments also metered input power to limit the difference between maximum chip temperature and fluid inlet temperature to 15 K. This power metering was not feasible for the current model. Instead, the model applied a constant heat flux of  $100 \text{ W/cm}^2$ . This constant heat flux resulted in a similar average temperature difference of 17.6 K. Although the difference varied from 44 K to 9.9 K across the flow rates.

### 3. Results and discussion

Results for the microchannel model are presented and then discussed with the same structure repeated for the manifold model.

#### 3.1. Microchannel model

Compared to the baseline geometry, the open geometry substantially reduced the pressure drop, with only a slight increase in thermal resistance. Fig. 6 plots these results across tested flow rates and channel widths.

Proportionate pressure drop reductions were higher at increased flow rates as expected due to a dependence on the square of the velocity. For example, at 1.2 L/min the pressure drop was reduced by an average of 69 % (9.4 kPa to 3.27 kPa) compared to 56 % (0.68 kPa to 0.33 kPa) at 0.2 L/min. Pressure drop reductions were also smaller for narrower channels, as channels contribute proportionately more to total pressure drop than the nozzles. For example, the  $100 \mu\text{m}$  wide channels had an average pressure reduction of 72 % compared to the  $25 \mu\text{m}$  wide channels at 53 %. Narrow channels exhibited the largest pressure drop reduction of 14.9 kPa to 6.4 kPa. Changes in thermal resistance were negligible ( $-0.6 \%$ ) for the narrowest channels. However, the thermal resistance increased slightly for the wider channels at higher flow rates. The  $100 \mu\text{m}$  wide channels at 1.0 L/min had the highest increase of 6.6 % or  $0.149 \text{ cm}^2\text{K/W}$  to  $0.159 \text{ cm}^2\text{K/W}$ . Jet flow the leading cause of these trends.

The baseline heat sink has microchannels which run across the entire chip. These uninterrupted microchannels are straightforward to manufacture and offer structural stability. However, the tops of the microchannel walls partially block the adjacent slot nozzles. They halve the cross-sectional flow area, acting as a nozzle tip. This small area causes the jet to accelerate, although with a significant pressure drop penalty as demonstrated in Figs. 7 and 8. The pressure drops about 2.0 kPa across the outlet constriction and 2.5 kPa across the inlet constriction which represents about 77 % of the total 5.8 kPa pressure drop. The open case mitigates this pressure drop by removing channel walls from underneath the slot nozzle. However, this does reduce the jet velocity.

Jet impingement is a well-established means to enhance local heat transfer rates. When a jet impacts, or impinges, on a surface the transfer of the jet's momentum suppresses the development of thermal and hydrodynamic boundary layers and creates a local increase in heat transfer rates. In both the baseline and open geometries, a narrow slot nozzle forms a jet which impinges onto the microchannels below. Fig. 8 shows the effect of this jet with the local heat transfer coefficient in the impingement area being more than twice that in the developed region:  $5.6 \text{ W/cm}^2\text{K}$  versus  $2.6 \text{ W/cm}^2\text{K}$ . This local increase in heat transfer rates is propor-

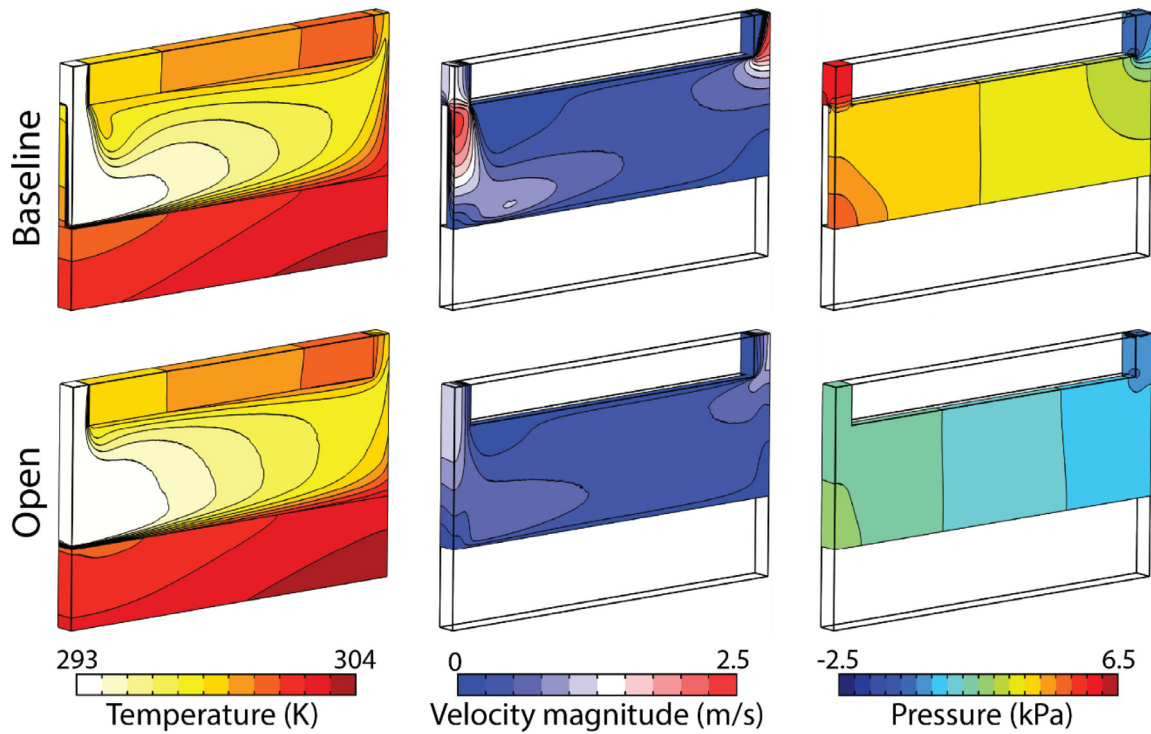


Fig. 7. Temperature, velocity magnitude and pressure contours for the baseline and open channel cases for  $w_{HT, ch} = 50 \mu m$  and  $V_{tot} = 1$  LPM

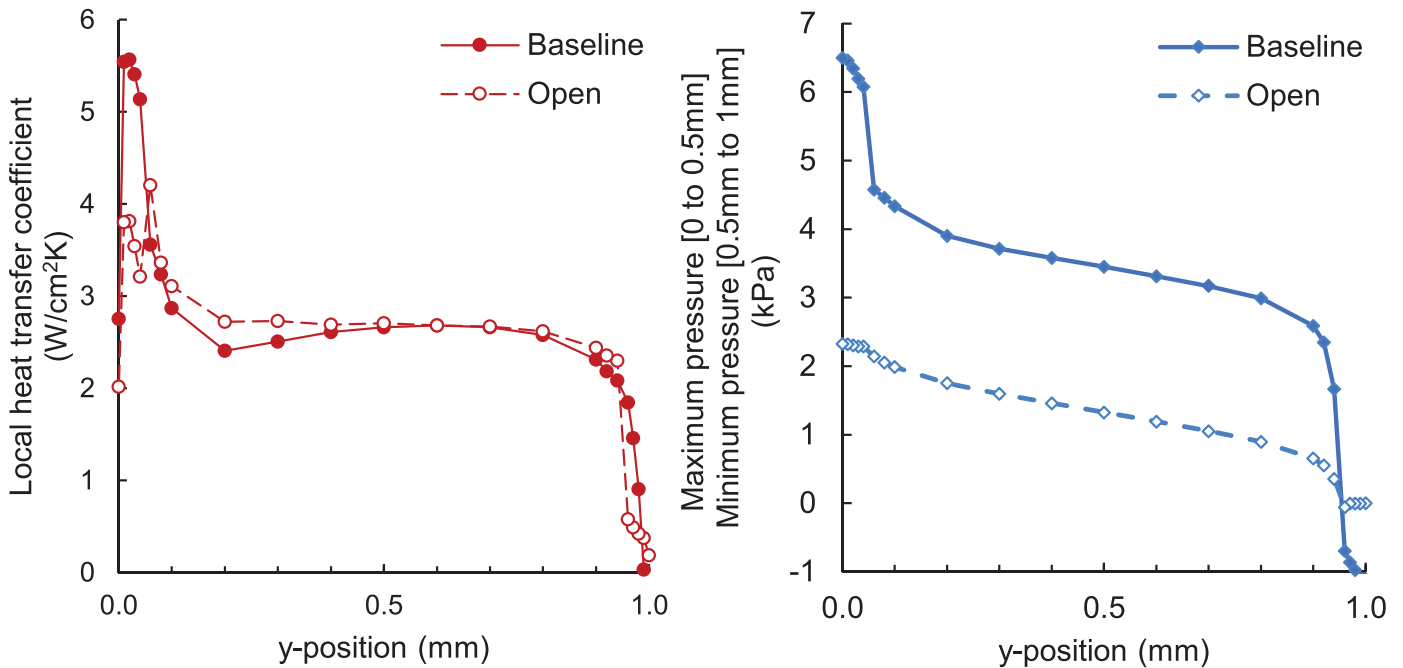


Fig. 8. Local values from microchannel inlet to outlet for  $w_{HT, ch} = 50 \mu m$  and  $V_{tot} = 1$  L/min. A) Local heat transfer coefficient calculated as in Escher, et al. [14] B) Maximum pressure for  $y = 0$  mm to 0.5 mm and minimum pressure for  $y = 0.5$  mm to 1 mm.

tional to the velocity of the jet. The baseline jet is about twice as fast as the open on (2.38 m/s vs. 1.25 m/s) which results in a 33 % greater local heat transfer coefficient in the impingement area. An associated issue, with the jet's high velocity, is the formation of a recirculation region near the trailing edge of the nozzle. The baseline local heat transfer coefficient is about 9 % lower in this region. The open case has a lower jet velocity and slightly increases the total thermal resistance. This suggests the local enhancements of the jet slightly outweigh its diminishments. However, the jet

flow is less influential for narrower channels or at lower flow rates.

Fig. 9 highlights the difference in jet flow for wide and narrow channels. For the wide channels, the jet impinges onto the bottom surface and forms a recirculation region near the inlet nozzle. This jet enhances and diminishes convective heat transfer rates in the respective impingement and recirculation regions. These jet effects on heat transfer are diminished for narrower channels because the high viscous drag from walls quickly dissipates the mo-

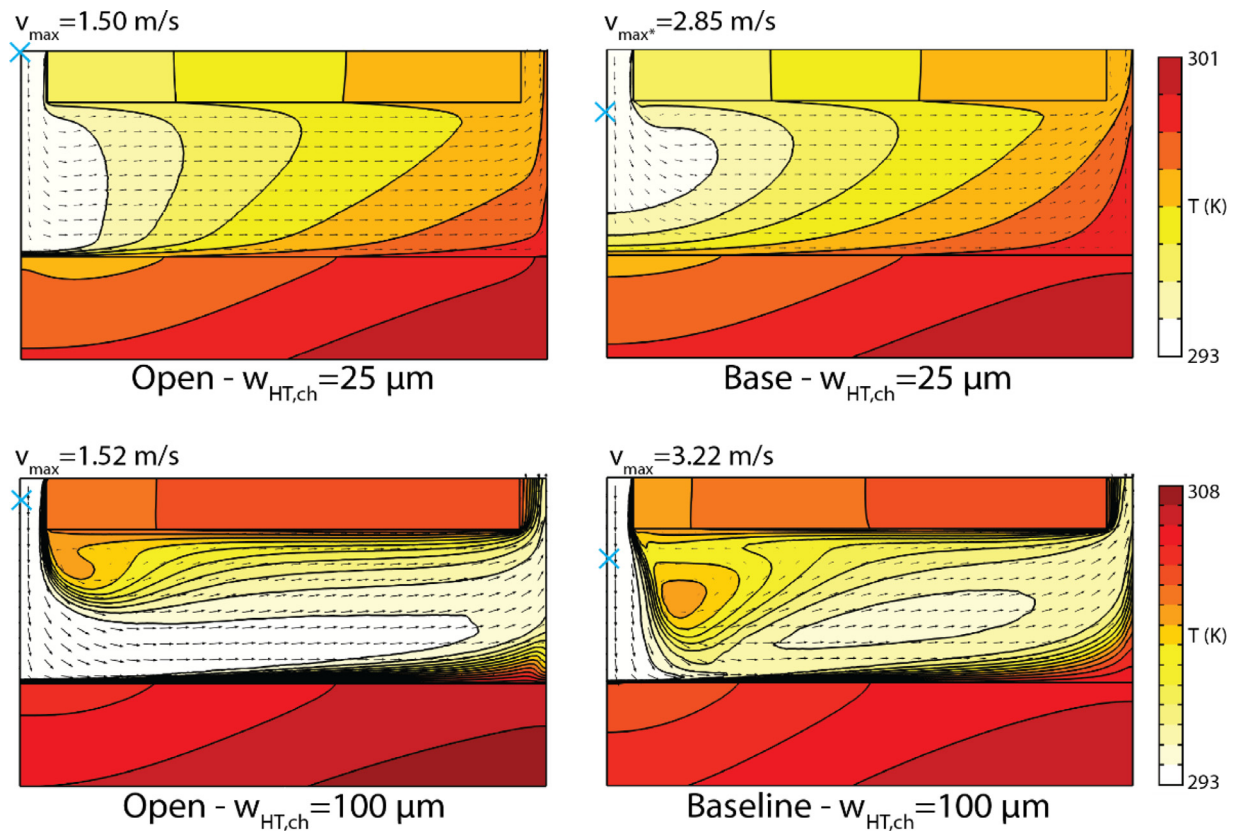


Fig. 9. Jet impingement for wide and narrow channels at high flow rates ( $V_{tot}=1.2$  L/min) Temperature contours with velocity vectors and peak velocities marked.

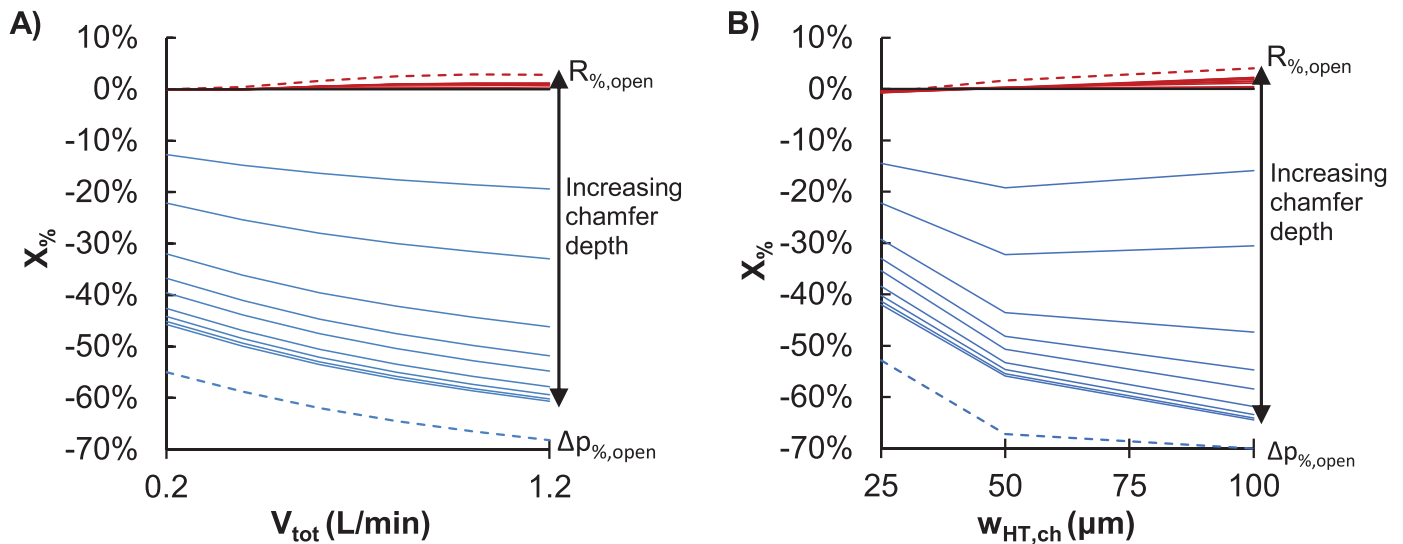


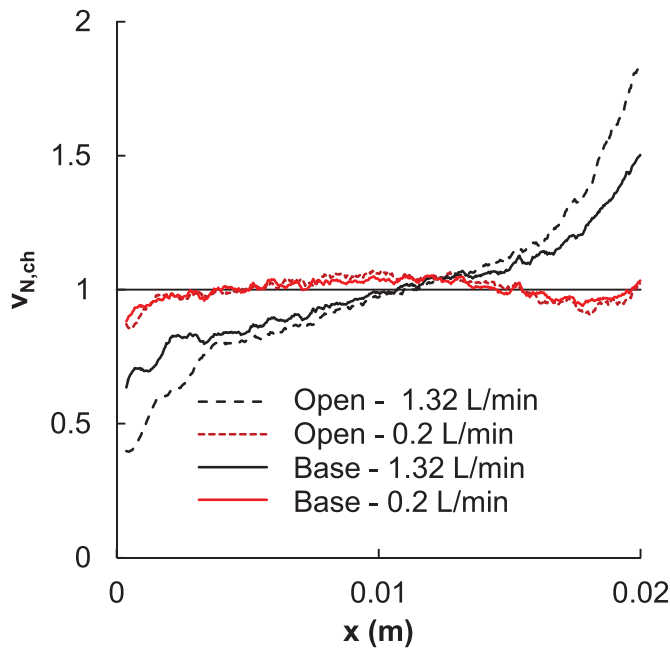
Fig. 10. Proportional reduction in thermal resistance (red) and pressure drop (blue) for the chamfered cases compared to the baseline. Chamfer depths are: (12.5, 25, 50, 75, 100, 150, 200, 250 and 300)  $\mu\text{m}$ . A) averages values over the flow rates while B) averages values over the channel widths.

mentum of the jet. As such, the open geometries reduction of jet velocity does not significantly influence heat transfer rates for the narrower channels.

The narrow channels high flow resistance quickly dissipates the momentum of the jet. Because of this, the reduced jet velocity does not significantly diminish total thermal resistance.

The chamfered case showed similar benefits to the open case, although, to a lesser degree. Fig. 10 compares the pressure drop and thermal resistance results. The magnitude of the changes is

proportional to the chamfer depth. For example, at a flow rate of 1 L/min, a channel width of 50  $\mu\text{m}$  and a chamfer depth of 25  $\mu\text{m}$  the pressure drop was reduced by 38 % (5.8 kPa to 3.6 kPa) whereas a chamfer 12  $\times$  deeper (300  $\mu\text{m}$ ) reduced the pressure drop by 62 % (5.3 kPa to 2.2 kPa). These results highlight the importance of the constriction at the nozzle tip to pressure drop. The removal of the nozzles square edge may have also contributed slightly. Unlike the open case, the chamfer design used a non-planar geometry. This geometry may be more



**Fig. 11.** Moving average of microchannel velocities. Velocities evaluated at the centre point of each channel and normalised with the average across all channels.

difficult to manufacture with conventional microfabrication techniques. However, it would offer superior structural stability to the open case, while retaining most of the pressure drop reduction benefits.

Using wider or shorter nozzles may also reduce the total pressure drop. However, these present some limitations compared to the open case. At a channel width of  $50\ \mu\text{m}$  and a flow rate of  $1\ \text{L/min}$ , doubling the nozzle width ( $w_{HT, N} = 200\ \mu\text{m}$ ) decreases the pressure drop by 54 %. This reduction is slightly less than the open reduction of 70 %, under the same conditions. Wider nozzles increase the minimum width of the manifold channel, so can be restrictive for converging manifold channels which rely on narrow passages. Wider nozzles do not restrict height tapered manifolds. However, height tapering may be difficult to construct with conventional microfabrication techniques. Also, thin heat sink profiles may restrict the use of height tapered manifolds. Under the same conditions, halving the nozzle height to  $50\ \mu\text{m}$  would decrease the pressure drop slightly, 6 %. However, the height of the nozzle is the same as the thickness of the manifold base, which is structurally critical. As such, if structurally suitable shorter nozzles are preferred.

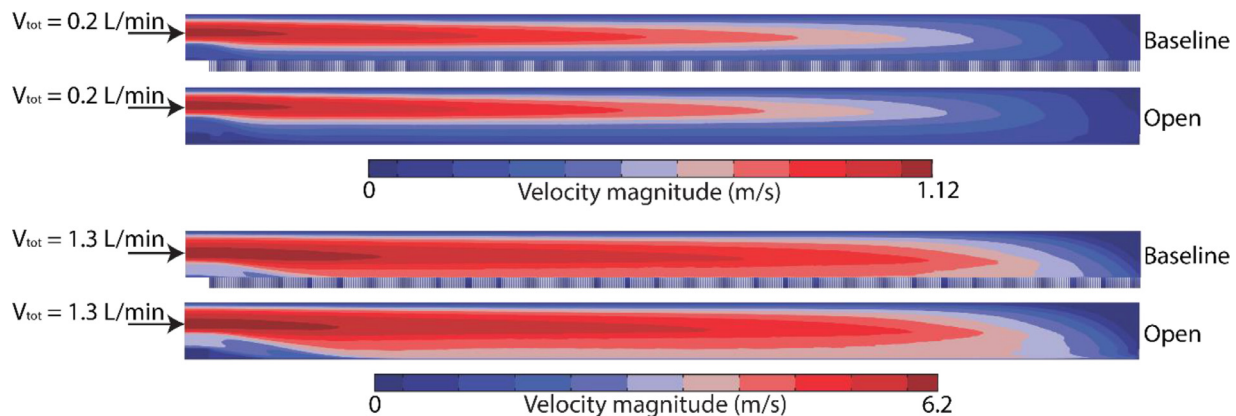
### 3.2. Manifold model

The manifold model provides insights on the heat sinks overall performance, not captured by the microchannel model. The microchannel model assumes each channel receives a uniform volume flux and at a uniform temperature. In practice, these conditions can be difficult to achieve. However, they can have a significant influence on performance. The microchannel model also assumes a vertical inlet flow direction. However, the flow has a significant horizontal component. The microchannel model also only captures the contribution of the heat transfer layer to the overall heat sink. Whereas, the manifold model captures the contribution of both layers. These features are now discussed, as well as overall performance results.

The convergence-divergence of the manifold channels is critical for achieving uniform flow rates across the microchannels. The current manifold model fully resolves hundreds of microchannels, which is computationally demanding. Consequently, an iterative optimisation procedure, for the manifold profile, is not feasible within the scope of this study. This study instead adopts the profiles optimised by Escher et al. [3,14]. As a result, the baseline case is better suited to the convergence-divergence than the open case. This discrepancy is most prevalent at higher flow rates, towards the start and end of the manifold channel. Fig. 11 plots the microchannel flow distribution more clearly. At a flow rate of  $1.32\ \text{L/min}$ , the baseline had normalised velocities between 1.51 and 0.55, whereas the open velocities had velocities between 0.40 and 1.86.

Recall, the open case removes channel walls from beneath the slot nozzles. This removal increases the cross-sectional area of the manifold channels. The increased flow area allows fluid to reach the ends of the channel with less resistance. Conversely, it promotes the formation of a broader stagnation region near the manifold's inlet. The velocity contours in Fig. 12 highlight these flow features. At the lower flow rates, the velocity distribution in the baseline and open geometries is similar. At higher flow rates, the broader stagnation region is observable. This stagnation region increases the thermal resistance of the open case at high flow rates.

Flow maldistribution degrades the heat sinks thermal performance, as highlighted in Fig. 13, which plots temperature along the base of the heat sink. These temperatures correspond to the flow distributions. The open case has slightly lower temperatures near the end of the channel,  $x > 10\ \text{mm}$ . Along the middle of the channel, the baseline and open temperatures are similar,  $3\ \text{mm} < x < 15\ \text{mm}$ . Near the inlet of the manifold channel,  $x < 3\ \text{mm}$ , the temperatures are similar for flow rates below  $0.45\ \text{L/min}$ . At higher flow rates, the open temperatures are much higher due to the formation of a broader stagnation region. The maximum chip temper-



**Fig. 12.** Velocity contours for baseline and open cases at  $0.2\ \text{L/min}$  and  $1.32\ \text{L/min}$ .

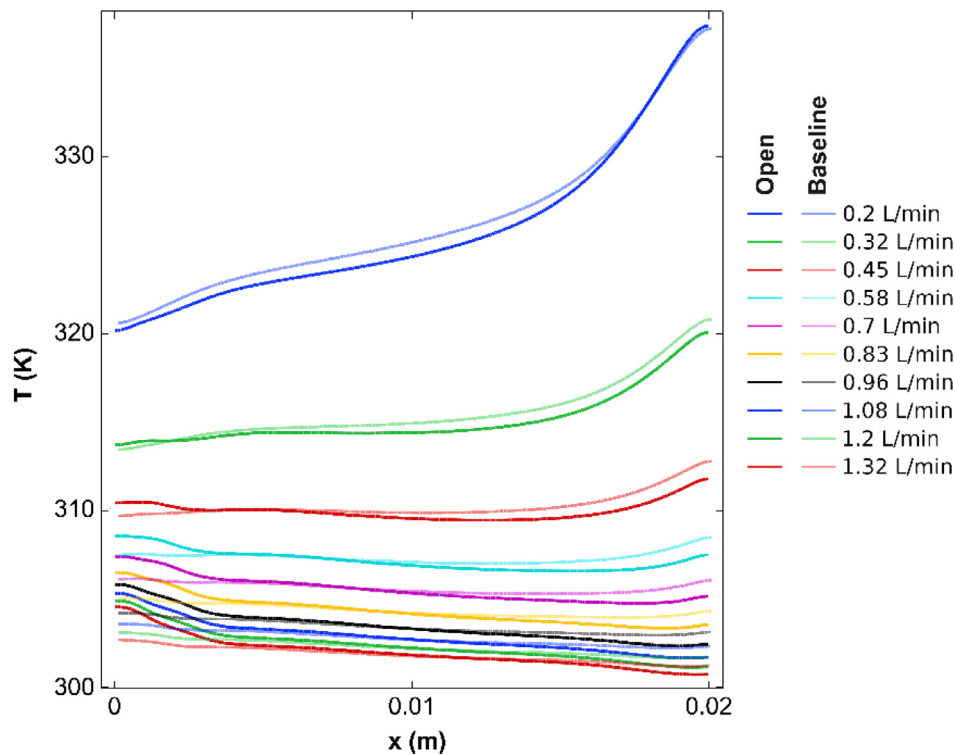


Fig. 13. Temperature evaluated at a cut line along the base of the heat sink:  $z = 0$  mm,  $y = 0.4$  mm.

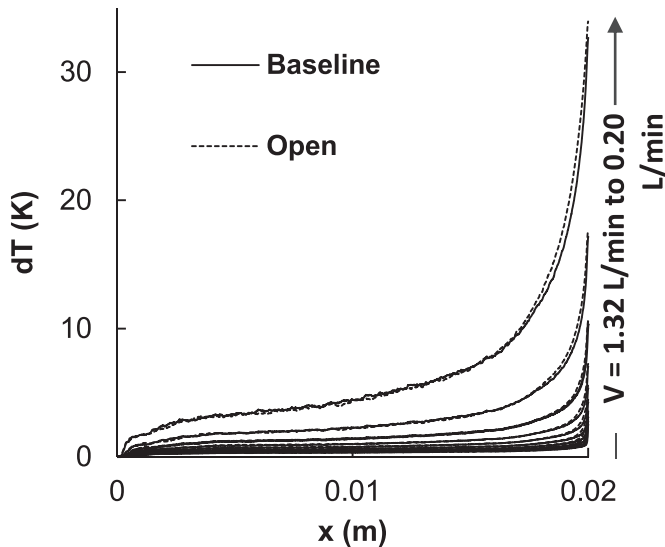


Fig. 14. Fluid temperatures evaluated along a cut line at the microchannel inlets:  $z = 0.851$  mm,  $y = 0$  mm.

ature occurs in this stagnation region. Consequently, the stagnation region limits the thermal resistance of the open case.

There is not a significant difference in fluid temperature for the baseline and open case. Fig. 14 plots the fluid temperature just above the microchannel inlets. As expected, both cases exhibit increasing temperatures in the streamwise direction. Streamwise heating is substantial for lower flow rates and small for higher flow rates. For lower flow rates, streamwise heating causes the heat sinks maximum temperature. Fig. 15 shows the maximum temperature towards the end of the channel. To counteract streamwise heating manifold profiles should promote streamwise increases in

volume flux, at lower flow rates. Streamwise heating is small at high flow rates. Hence, uniform flow distributions are still preferred.

Fig. 16 compares the total pressure drop and thermal resistance for the baseline and open cases. The open case reduces the pressure drop by 0.23 kPa to 3.40 kPa or about 25 %. At flow rates below 0.58 L/min, thermal resistance is within -4 % to 2 %. At flow rates above 0.7 L/min thermal resistance increase by 11 % to 21 %. Different flow distributions are the primary cause of differences in thermal resistance. The inlet stagnation region is particularly detrimental to open channel performance. Optimisation of the manifold profile may mitigate this issue while retaining pressure drop reductions. The open pressure drop reductions are primarily due to less constriction and expansion at the nozzle tip.

The manifold model shows a significant lengthways velocity at the microchannel inlet. However, the microchannel model does not account for this lengthways component. The red regions in Fig. 18 help highlight the high lengthways velocity regions. Note, lengthways refers to the x-component, while vertical refers to the z-component. Fig. 17A plots the flow direction just above the microchannel inlets, where  $0^\circ$  is lengthways, and  $90^\circ$  is vertical. Here, the flow direction varies between about  $10^\circ$  to  $30^\circ$ . Predominantly lengthways. As the flow enters channels, it sheds this lengthways momentum as it impinges on the far channel wall. For the baseline case, the impingement occurs at the tip of the channel wall. For the open case, the impingement occurs across the height of the channel wall. Fig. 17B visualises this impingement for the baseline, by plotting arrow vectors in the region. The velocity vectors show a small recirculation region and a quick elimination of the lengthways velocity component. The effects of this lengthways impingement do not significantly affect the performance of these narrow microchannels. However, wider channels may be more susceptible to this impingement.

The baseline and open cases show vertical jet impingement, onto the base of the microchannels. The baseline case shows

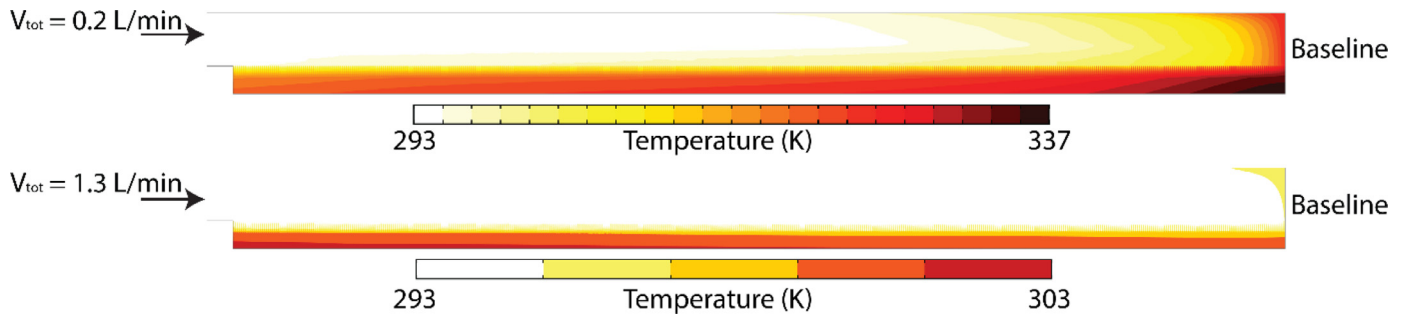


Fig. 15. Temperature contours evaluated along inlet manifold:  $y = 0$  mm,  $x = 0$  mm.

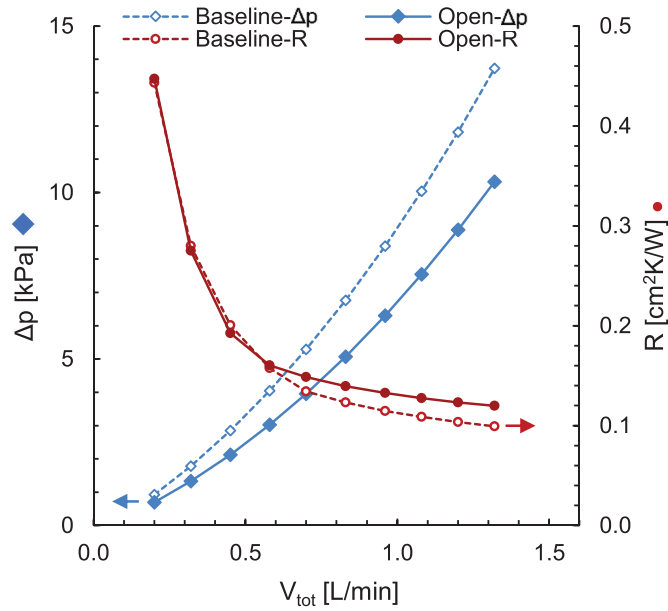


Fig. 16. Pressure drop and thermal resistance for the baseline and open cases.

higher downwards jet velocities than the open case, aligning with results from the microchannel model. For example, at a flow rate of 1.32 L/min, the downwards velocity peaks at 1.54 m/s and 0.89 m/s for the baseline and open case, respectively. The open case shows lower temperatures at the microchannel entrance, due to the exposure of the channel sidewalls to the high lengthways velocity. Compared to flow distribution, these local effects are not significant for overall performance. Note, elevated pressures in the impingement regions are visible in Fig. 18.

Results from the manifold model support the adiabatic boundary condition used the microchannel model, for higher flow rates. Fig. 19 plots the mass averaged fluid temperature rise across the microchannels, as a proportion of the total rise. For the baseline case at 1.3 L/min, about 88 % of the temperature increase occurs across the microchannels. Compared to 58 % at 0.2 L/min. This temperature rise does not include the contribution of the slot nozzles. The adiabatic tip condition is reasonable when investigating microchannel performance. However, it will lead to an underestimation of thermal resistance. Optimisation studies should account for the contribution for more accurate results, particularly at lower flow rates. In Fig. 19, the microchannel contribution is larger for the open case. This larger contribution is due to the position of temperature calculation, the plane is slightly offset from the mi-

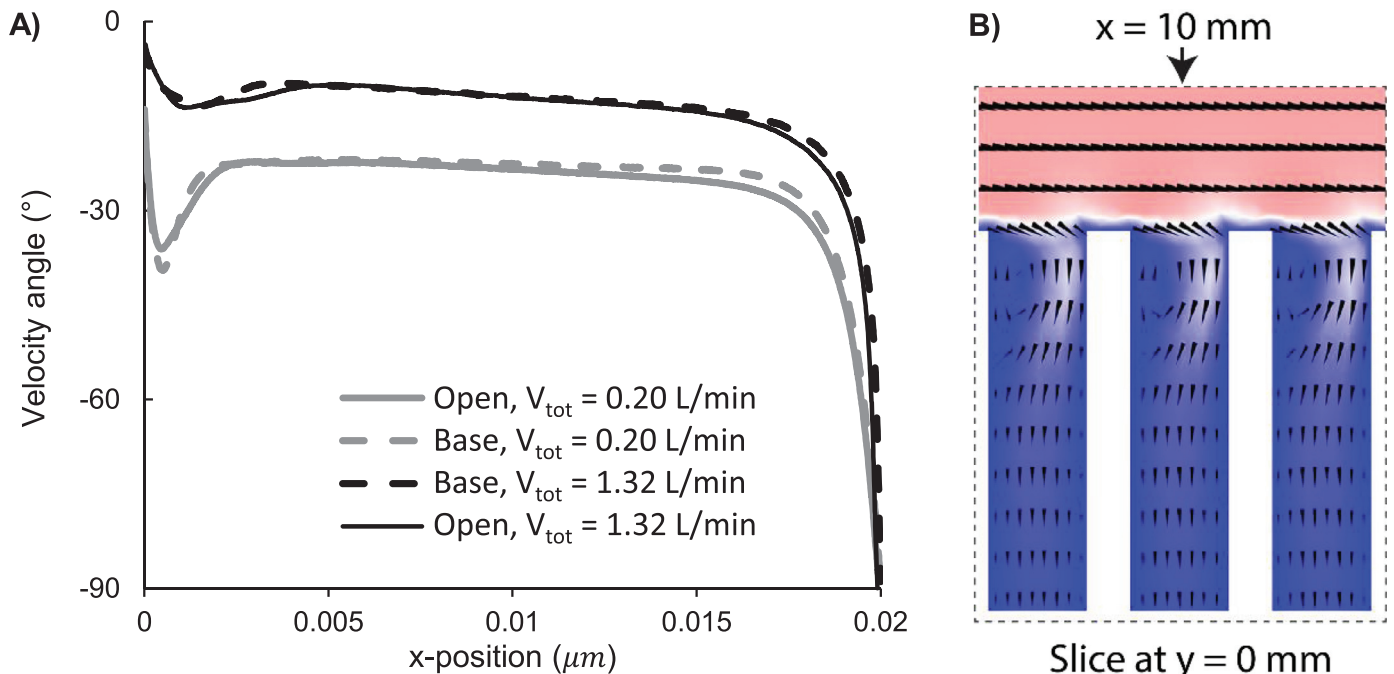


Fig. 17. A) The direction of velocity where  $0^\circ$  is lengthways (x-axis) and  $-90^\circ$  is vertically down (z-axis). Evaluated at cut line near microchannel inlet:  $z = 0.851$  mm,  $y = 0$  mm. B) Velocity vectors showing the flow path into the three middle microchannels:  $x = 9.92$  mm to  $10.79$  mm,  $y = 0$  mm.

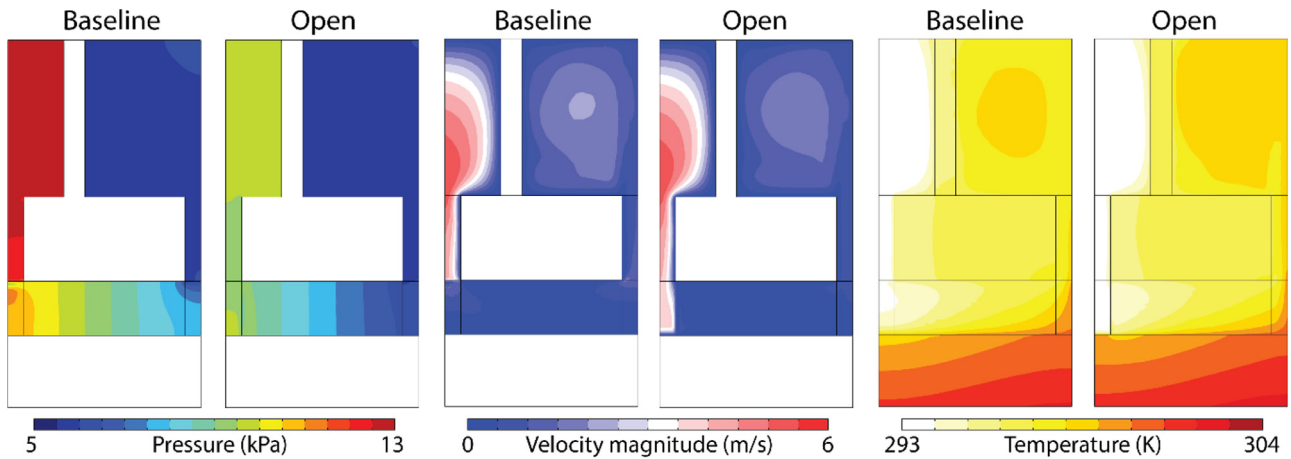


Fig. 18. Pressure, velocity and temperature contours at the middle channel for baseline and open cases. Evaluated at yz cut plane,  $x = 10$  mm.

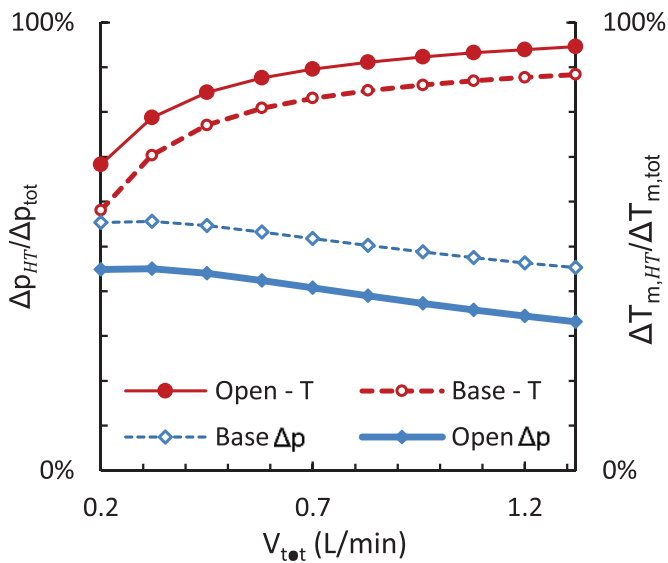


Fig. 19. Fluid temperature rise and pressure drop across the microchannels as a portion of the total. The total evaluated across manifold inlet and outlet. Microchannel evaluated at microchannel inlet and outlet, where  $z = 0.751$  mm.

crochannel entrance, whereas it intersects for the baseline case. This intersection of the microchannel tips leads to slightly higher temperature due to the proximity to channels. Interestingly the fluid temperature slightly decreases,  $< 0.5$  K, after it exits the microchannels. This due to heat transfer across the manifold wall, to the cooler fluid in the inlet manifold.

Fig. 19 also plots the contribution of the microchannel to the total pressure drop. Across the flow rates, about half the pressure drop occurs in the microchannels. The remainder of the pressure drop occurs in the manifold channels and slot nozzles. The significant contribution of manifold channels to total pressure drop highlights a limitation of the microchannel model, which only determines pressure drop across the slot nozzles and microchannels. For conventional geometries, analytical methods or simulations with representative porous mediums present an adequate solution, however, novel geometries may require fully resolved models.

#### 4. Conclusion and future work

The study proposed an open manifold microchannel heat sink to meet the intensifying demands of electronics cooling. The open

geometry substantially reduced the pressure drop of a conventional heat sink by eliminating ineffective nozzle constrictions, specifically, by removing microchannel walls that usually lie directly beneath slot nozzles. The open geometry also retained the benefits of a conventional manifold microchannel heat sink: low thermal resistance, thin profile, liquid waste heat recovery and a simple planar geometry well suited to microfabrication.

Two three-dimensional conjugate heat transfer models were used: encompassing a single microchannel, and a single manifold channel pair including the hundreds of interconnecting microchannels. Results were then compared to an unmodified, or baseline, manifold microchannel heat sink geometry.

The open geometry reduced microchannel pressure drop by 57 % (14.9 kPa to 6.4 kPa) for narrow channels (25  $\mu$ m) at a high flow rate (1.2 L/min) with a negligible change to thermal resistance (0.0782  $\text{cm}^2\text{K/W}$  to 0.0780  $\text{cm}^2\text{K/W}$ ). Wider channels (100  $\mu$ m) increased the microchannel pressure drop reduction to 77 % (5.4 kPa to 1.22 kPa), although with a slight increase in thermal resistance 5.5 % (0.1378  $\text{cm}^2\text{K/W}$  to 0.1454  $\text{cm}^2\text{K/W}$ ). The increased thermal resistance was due to the reduced jet velocity. Different jet velocities did not significantly influence heat transfer rates for narrow channels as high viscous drag quickly dissipated the momentum of the jet. Chamfering the microchannel tips near the slot nozzle showed similar benefits to opening them, although to a lesser degree.

The open geometry also reduced the total heat sink pressure drop by 25 % (13.7 kPa to 10.3 kPa) at a high flow rate (1.32 L/min). At lower flow rates ( $< 0.58$  L/min), the thermal resistance decreased slightly ( $\sim 4$  % to 1 %). At higher flow rates ( $> 0.70$  L/min), the thermal resistance increased moderately (11 % to 21 %). Both changes in thermal resistance were due to flow distribution across the microchannels. In particular, the increase was caused by a stagnation region near the manifold inlet, as opposed to diminished microchannel heat transfer rates. Dedicated optimisation of the manifold convergence profile could mitigate this maldistribution. The lowest thermal resistance for the baseline and open geometries were 0.099  $\text{cm}^2\text{K/W}$  and 0.120  $\text{cm}^2\text{K/W}$ , respectively. The manifold model also showed a substantial horizontal velocity component at the microchannel inlets. Microchannel models usually assume the flow to be vertical. Additionally, the manifold channels were shown to have a significant contribution to overall heat transfer rates (68 % to 95 %).

Uniform flow distribution to individual microchannels is crucial to achieving the ideal thermal performance of manifold microchannel heat sinks. As such, there is a need for robust methods for achieving uniformity, as existing methods usually require

a dedicated optimisation procedure. Larger cooling areas increase the challenge. Experimental tests are also required to affirm the structural stability of the open geometry. Future studies may also utilise the fully resolved manifold model to investigate flow behaviour or other novel geometries, such as those which leverage the significant horizontal velocity component. Finally, future studies may also test the current model under a broader range of conditions: coolant types, elevated fluid temperature or other geometric variations.

## Author Statement

**Nicholas Gilmore:** Conceptualization, Methodology, Software, Formal analysis, Validation, Writing – Original, Draft, Writing - Review & Editing, Visualization

**Victoria Timchenko:** Supervision, Writing - Review & Editing

**Chris Menictas:** Conceptualization, Resources, Supervision, Writing - Review & Editing, Resources, Project administration

## Declaration of Competing Interest

none

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